



Inheritance

Introduction

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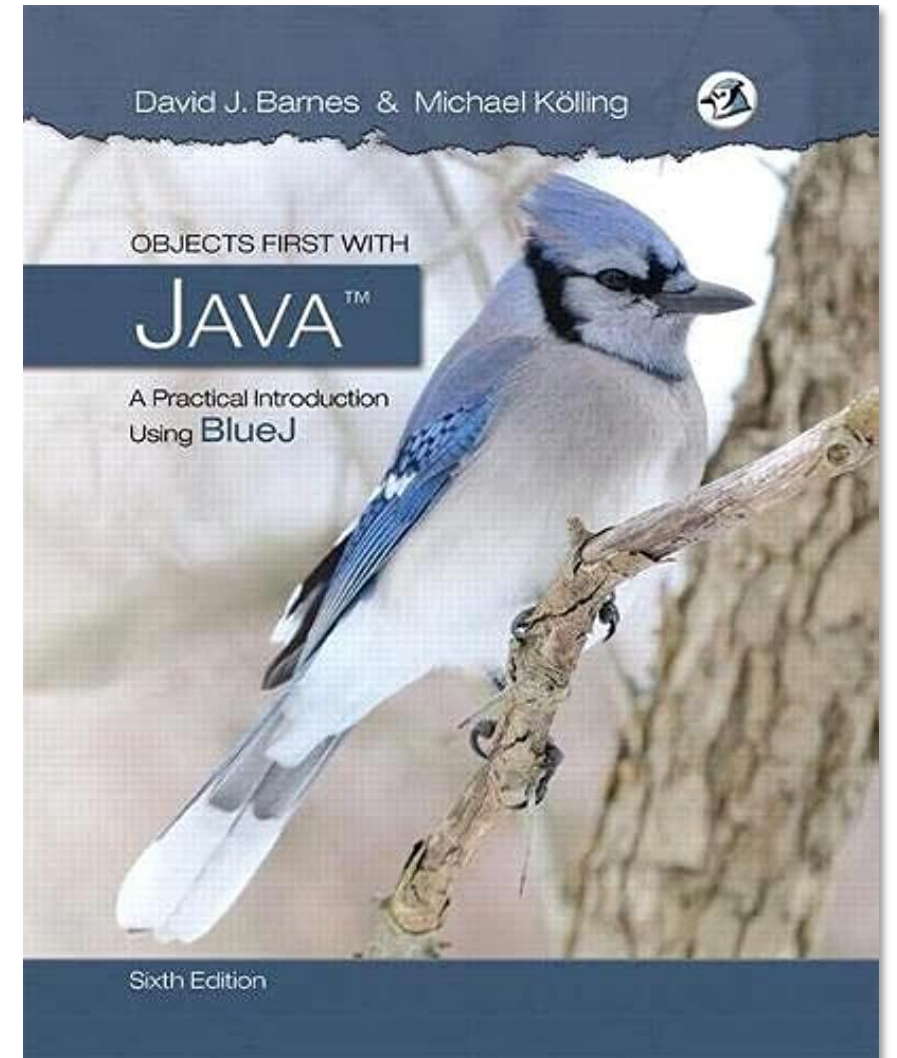
Agenda

- Inheritance Hierarchies
- Social Network v5
- Sub and Super Classes



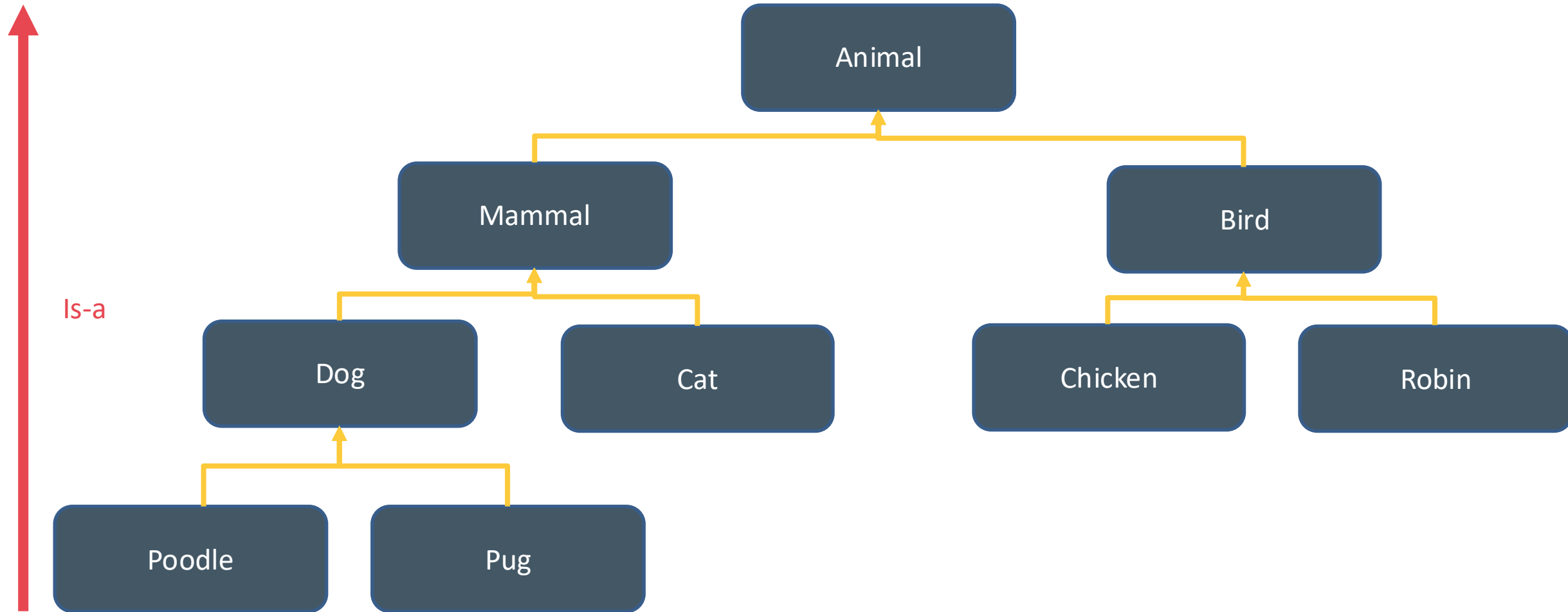
Lectures & Labs

- This weeks lectures and labs are based on examples in:
- **Objects First with Java**
 - A Practical Introduction using BlueJ, © *David J. Barnes, Michael Kölling*

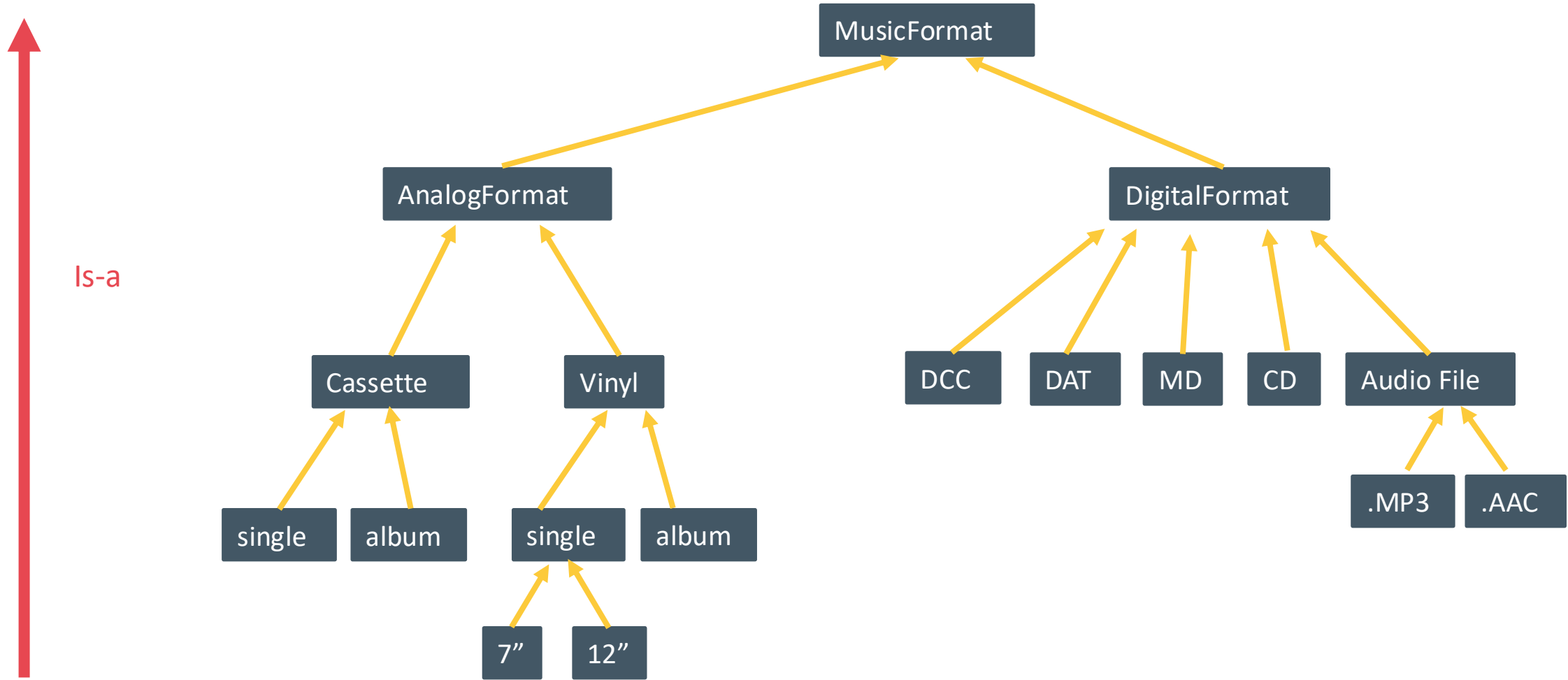


Inheritance Hierarchies

INHERITANCE HIERARCHIES

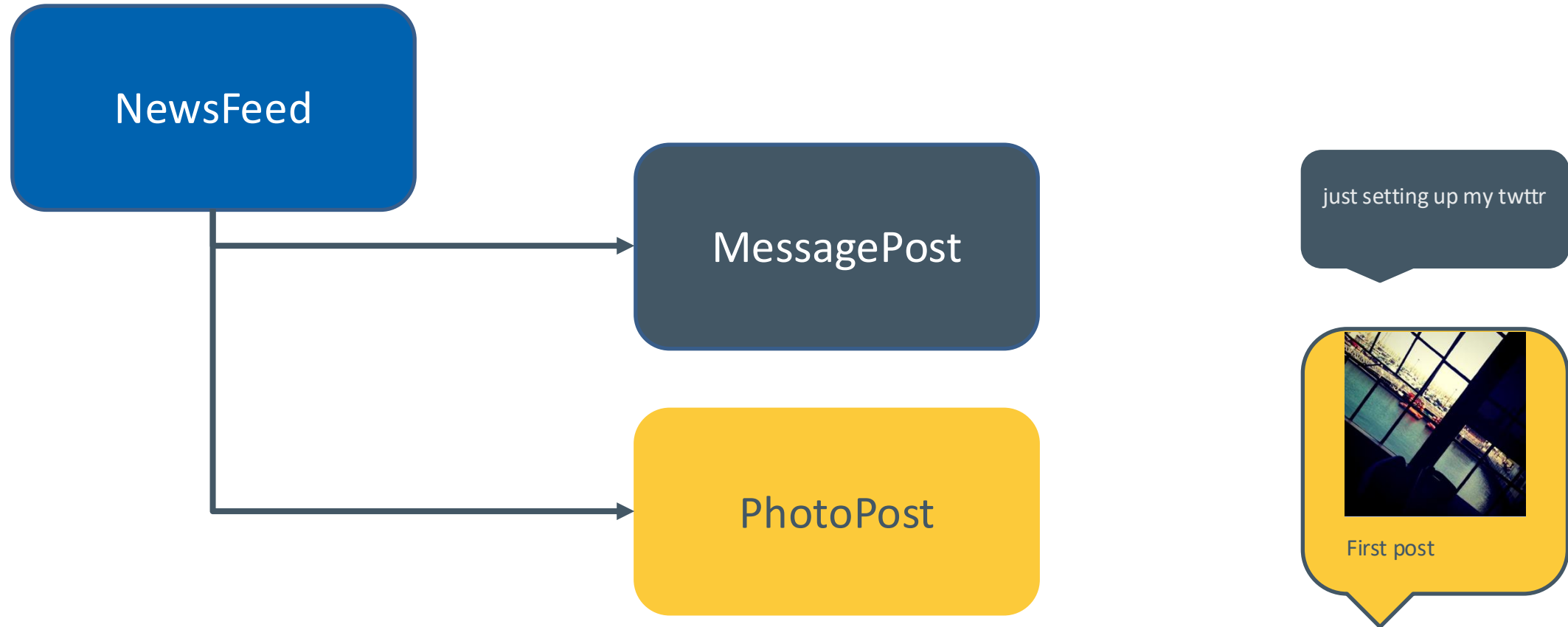


INHERITANCE HIERARCHIES



Social Network v5

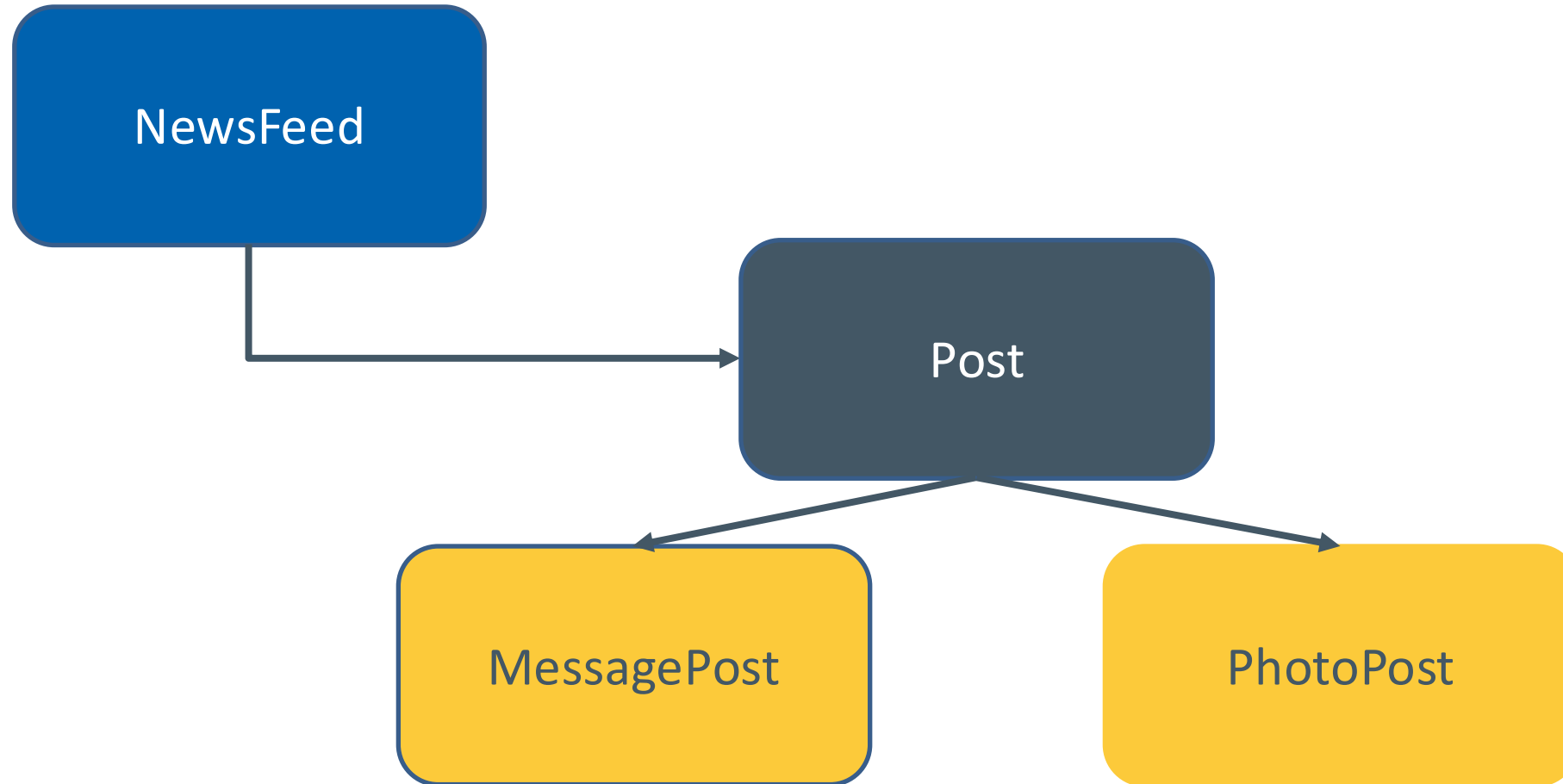
Social Network v4 Recap – Class Diagram



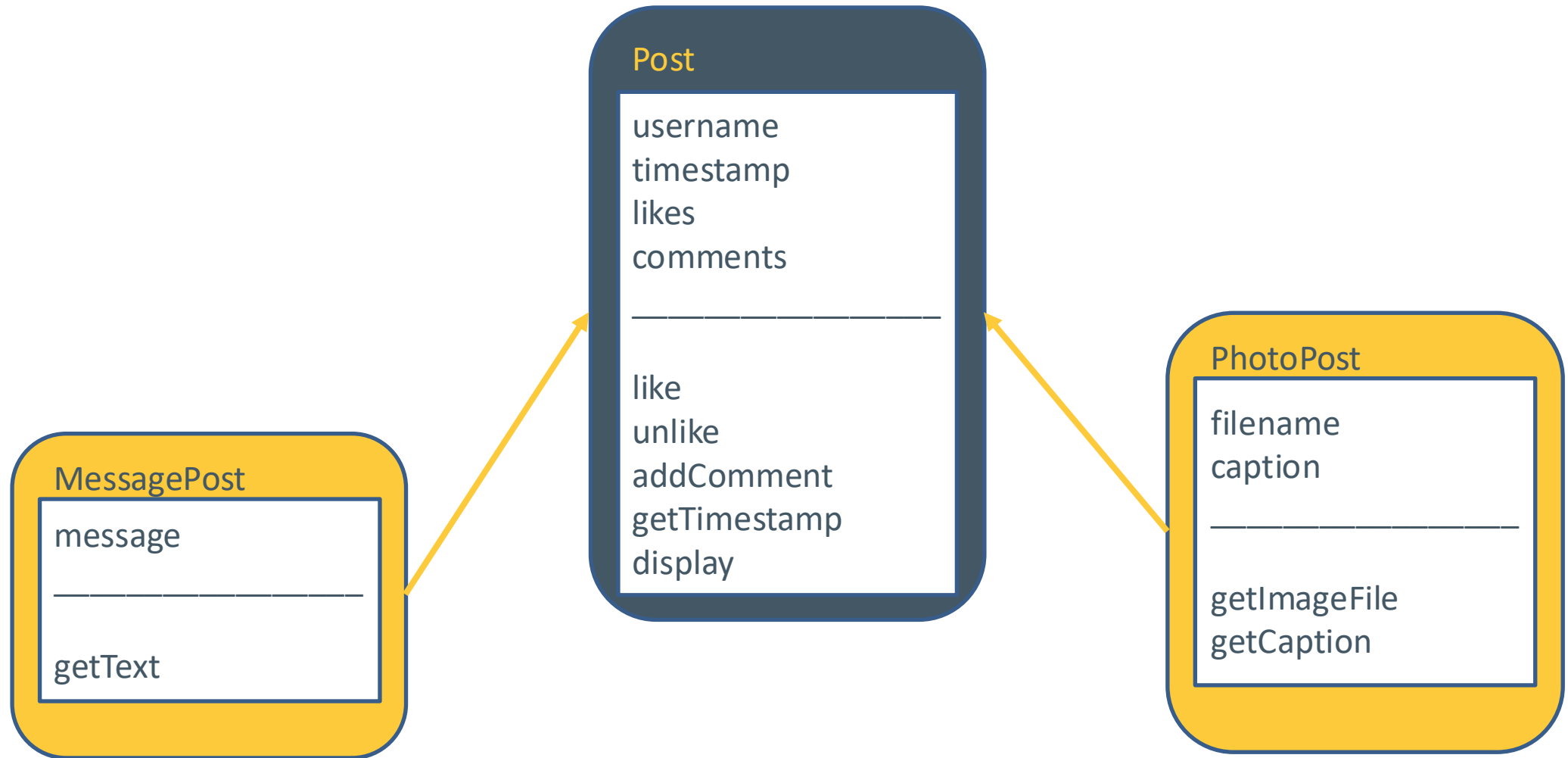
CRITIQUE OF SOCIAL NETWORK V4

- Code duplication:
 - **MessagePost** and **PhotoPost** classes very similar (large parts are identical)
 - makes maintenance difficult/more work
 - introduces danger of bugs through incorrect maintenance
- Code duplication in NewsFeed class as well.

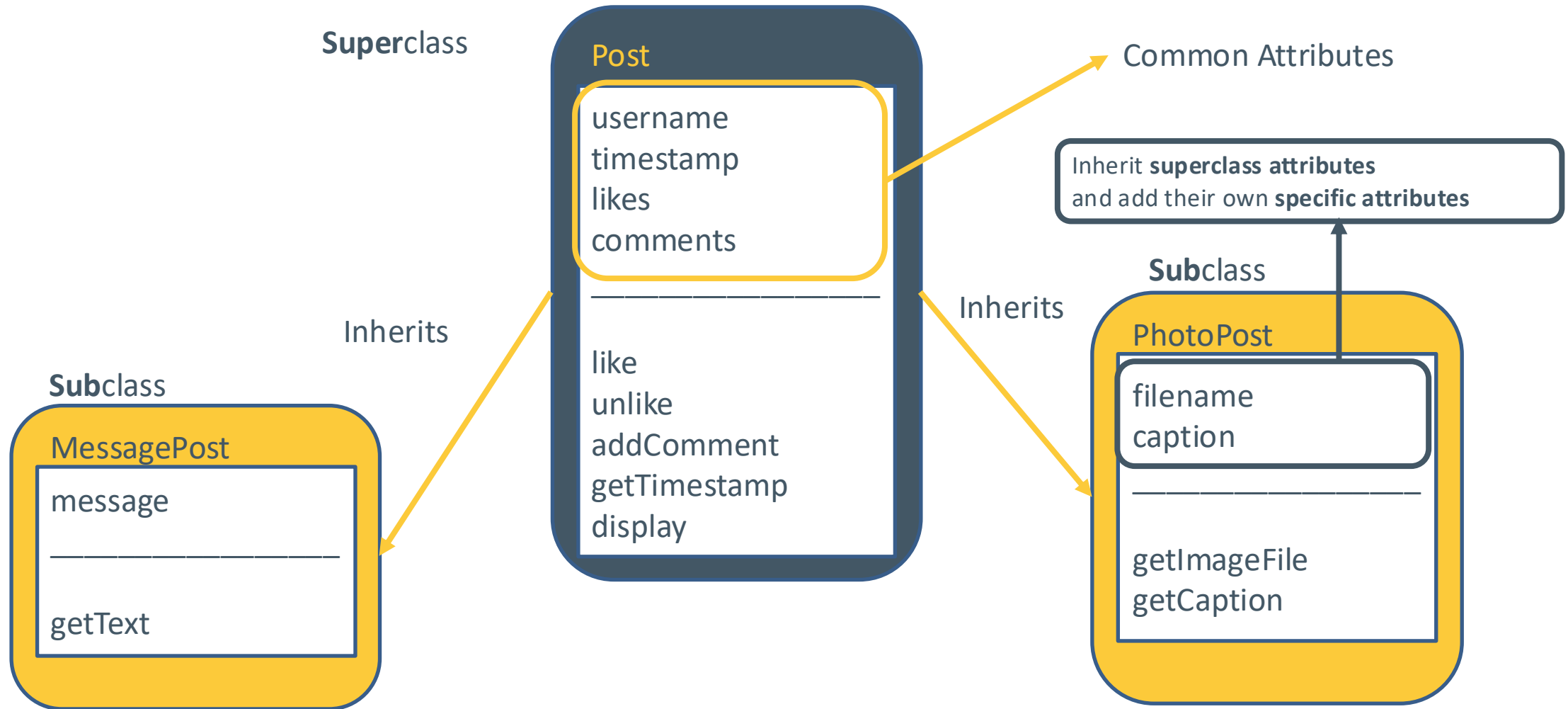
Social Network v5 Class Diagram



SOCIAL NETWORK V5 - CLASSES



SOCIAL NETWORK V5 - CLASSES

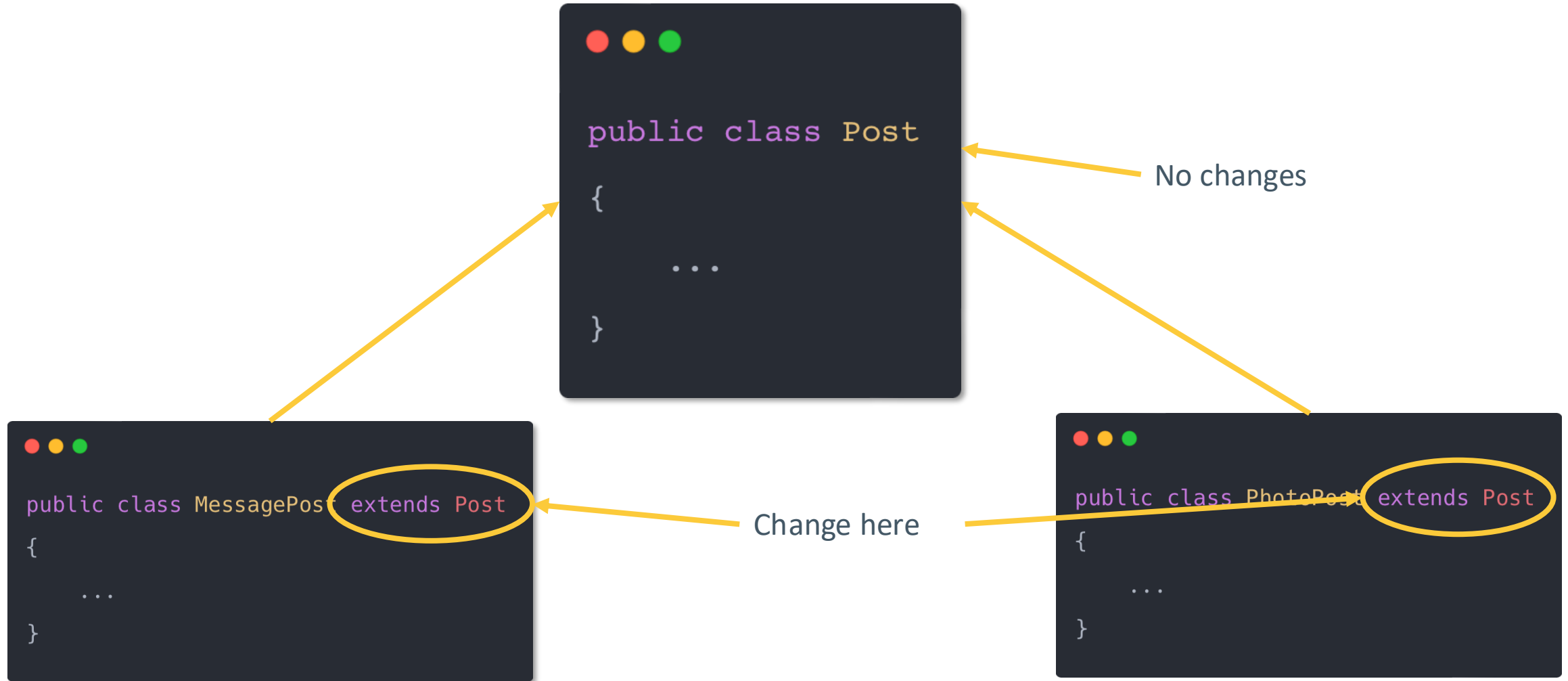


SOCIAL NETWORK V5 – INHERITANCE SUMMARY

- define one superclass
 - Post
- define subclasses for
 - MessagePost
 - PhotoPost
- the superclass
 - defines common attributes (via fields)
- the subclasses
 - inherit the superclass attributes (fields)
 - add other specific attributes (fields)

Super and Sub classes – code

INHERITANCE IN JAVA



SUPERCLASS

Common Fields

```
public class Post
{
    private String username;
    private long timestamp;
    private int likes;
    private ArrayList<String> comments;

    // constructor and methods omitted.
}
```

SUBCLASSES

```
public class PhotoPost extends Post
{
    private String filename;
    private String caption;

    // constructor and methods omitted.
}
```

```
public class MessagePost extends Post
{
    private String message;

    // constructor and methods omitted.
}
```

Subclass fields are unique to that subclass

Subclass objects inherit all fields from the superclass

INHERITANCE & CONSTRUCTORS - SUPERCLASS

Nothing unusual in the superclass

```
public class Post
{
    private String username;
    private long timestamp;
    private int likes;
    private ArrayList<String> comments;

    /**
     * Initialise the fields of the post.
     */
    public Post(String author)
    {
        username = author;
        timestamp = System.currentTimeMillis();
        likes = 0;
        comments = new ArrayList<String>();
    }

    // methods omitted
}
```

INHERITANCE & CONSTRUCTORS - SUBCLASS

- Must extend the superclass
- Must call the superclass constructor

```
public class MessagePost extends Post
{
    private String message;

    /**
     * Constructor for objects of class MessagePost
     */
    public MessagePost (String author, String text)
    {
        super(author);
        message = text;
    }

    // methods omitted
}
```

SUPERCLASS CONSTRUCTOR CALL

- Subclass constructors must always contain a **super** call.
- If none is written, the compiler inserts one (without parameters)
 - works only, if the superclass has a constructor without parameters
- **super** call must be the first statement in the subclass constructor.

Questions